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Video 4 · Geometry Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 16–19 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

4	14	Q16–Q19	14:50
questions	marks	E3.4 – E4.7	video runtime

M1 method mark A1 accuracy mark (needs the M mark) B1 independent mark

Question 16

Midpoint, length and perpendicular lines

$A(-2, 5)$ and $B(4, -3)$

(a) Find the coordinates of the midpoint of AB. [1]

Add the x's and halve, add the y's and halve:

$$\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right) = \left(\frac{2}{2}, \frac{2}{2}\right)$$

(1, 1) B1

(b) Find the length of AB. [2]

Change in x: $4 - (-2) = 6$ Change in y: $-3 - 5 = -8$

Pythagoras: $AB = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100}$ M1

10 A1

Tip: squaring removes the minus sign, so $(-8)^2 = 64$.

Question 16 (continued)

Midpoint, length and perpendicular lines

(c) Find the equation of the perpendicular bisector of AB, in the form $y = mx + c$. [3]

1. Gradient of AB = $\frac{\text{change in } y}{\text{change in } x} = \frac{-8}{6} = -\frac{4}{3}$ M1

2. Perpendicular gradient: flip and change sign → $\frac{3}{4}$ M1

3. It passes through the midpoint (1, 1) from part (a):

$1 = \frac{3}{4} \times 1 + c \rightarrow c = \frac{1}{4}$

$$y = \frac{3}{4}x + \frac{1}{4}$$ A1

"Perpendicular bisector" = cuts AB in half (so it goes through the **midpoint**) at 90° (so use the **perpendicular gradient**).

Check: $-\frac{4}{3} \times \frac{3}{4} = -1$ ✓

Common mistake: using point A or B instead of the midpoint.

Question 17

Angles in regular polygons

The interior angle of a regular polygon is 156°. Find the number of sides. [2]

1. Exterior angle = 180° − 156° = 24° M1

2. Exterior angles add up to 360°, so:

Number of sides = 360 ÷ 24

15 A1

Why exterior? It is much easier than the interior-angle formula. Interior + exterior = 180° on a straight line.

Quick division: 24 × 10 = 240, 24 × 5 = 120, and 240 + 120 = 360, so 15.

Common mistake: 360 ÷ 156. Always divide 360 by the **exterior** angle.

Question 18

Circle theorems

*Each mark needs the angle **and** the reason, written in the syllabus wording.*

(a) Find angle *AOC*. Give a reason. [1]

Angle *ABC* = 68° is at the circumference, standing on arc *AC*.

AOC = 2 × 68°

136° B1

Reason: the angle at the centre is twice the angle at the circumference.

(b) Find angle *ADC*. Give a reason. [1]

ABCD is a cyclic quadrilateral. *B* and *D* are opposite.

ADC = 180° − 68°

112° B1

Reason: opposite angles of a cyclic quadrilateral add up to 180°.

(c) Find angle *CAT*. Give a reason. [1]

TA is a tangent and *AC* is a chord.

The angle between them equals the angle in the **other** segment, which is *ABC*.

68° B1

Reason: alternate segment theorem.

Question 19

Similar solids — area and volume

Surface areas 45 cm² and 80 cm². Smaller volume 54 cm³. Find the larger volume. [3]

Area ratio: 45 : 80 = 9 : 16 (÷ 5)

Length ratio: square-root → 3 : 4, so scale factor $\frac{4}{3}$ M1

Volume ratio: cube → 27 : 64, so volume factor $\frac{64}{27}$

Larger volume = $54 \times \frac{64}{27} = 2 \times 64$ M1

128 cm³ A1

The rule: lengths k · areas k^2 · volumes k^3. Always go back to the **length** ratio first.

Common mistake: $54 \times \frac{80}{45} = 96$, using the area ratio for volume. This scores 0.

Non-calculator trick: 54 ÷ 27 = 2 first, then × 64.

Answers at a glance

Video 4 · Geometry · 14 marks

Q	Topic	Answers	Marks
16	Midpoint, length, perpendicular bisector	(a) (1, 1) (b) 10 (c) $y = \frac{3}{4}x + \frac{1}{4}$	6
17	Angles in regular polygons	15 sides	2
18	Circle theorems	(a) 136° (b) 112° (c) 68° (with reasons)	3
19	Similar solids	128 cm³	3
	Total		**14**

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. Next: **Video 5 · Mensuration, trigonometry, transformations and vectors (Q20–Q23)**.